KARTIK GOEL

CLOUD INFRASTRUCTURE & LINUX SYSTEMS · FULL-STACK DEVELOPMENT · ALGORITHMIC PROBLEM SOLVING

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SUMMARY

Computer Science student at JIIT with hands-on experience in full-stack development, systems programming, and cloud basics. Built real-world projects using React, Node.js, Python, and C++ — including a network traffic analyzer and an AI-assisted campus query system. Strong in data structures, algorithms, and problem-solving, with a consistent competitive programming track record. Comfortable working independently or in teams, eager to take on technically challenging work and grow under experienced engineers.

EDUCATION

Jaypee Institute of Information Technology (JIIT) — Expected May 2027
B.Tech – Computer Science Engineering | CGPA: 8.24 / 10 — *Noida, India*
Relevant Coursework: Operating Systems, Computer Networks, Distributed Systems, Open Source Software Development, Web Technology, Database Management Systems, Network Security, Software Engineering

SD Vidya Mandir, Panipat — 2023
Class 12 – CBSE | PCM Stream — *Panipat, India*
Overall: 92% | **PCM:** 96.67%

TECHNICAL SKILLS

Languages & Scripting: Python, Bash, JavaScript, TypeScript, C++, SQL, HTML5/CSS3

Databases: MongoDB, MySQL, PostgreSQL, Firebase, MongoDB Atlas — schema design & query optimisation

Cloud & Virtualization: AWS & GCP basics, Docker, Kubernetes basics, Virtualization principles

Automation & CI/CD: Bash scripting for health-checks & monitoring, CI/CD (GitHub Actions, Vercel)

Backend / Full-Stack: Node.js, Express.js, React.js, Next.js, Redux, Socket.io, REST API design, JWT Auth

AI / ML: LLM integrations (OpenAI, Gemini, Groq), RAG pipelines, FAISS, BM25, NumPy, Scikit-learn, Pandas

Networking: IP subnetting, VLANs, DNS, routing basics, load-balancing concepts

Tools & Collaboration: Git, GitHub, Vercel, Netlify, Streamlit, Render

PROJECT EXPERIENCE

JIIT Bot – Intelligent Campus Query System (RAG + Hybrid Search) — Nov 2025 – Present
Tech Stack: Python, Streamlit, FAISS, BM25, BeautifulSoup, Sentence Transformers, Gemini AI, Groq, OpenAI | **Live:** jiitbot-assistant.streamlit.app | **Source:** github.com/goyalkartik773/JiitBot

- Built a full data ingestion pipeline using BeautifulSoup4 to crawl JIIT's sitemap.xml, extracting 1000+ pages and PDFs into a structured document store, then indexed them using FAISS (384-dim embeddings) + BM25 fused via reciprocal rank fusion for hybrid semantic-keyword search.
- Integrated multi-LLM backend routing (Groq Llama-3.3-70B primary, OpenAI GPT-4o-mini fallback) with context-aware prompting, delivering cited, hallucination-resistant answers grounded in official JIIT sources.
- Implemented source citation in every response — all answers are grounded in official JIIT web pages and PDFs, with automatic offline fallback mode when LLM APIs are unavailable.

EatHub – Real-Time Food Delivery Platform — Sep 2025 – Present
Tech Stack: React (Vite), Node.js, Express.js, MongoDB, Firebase Auth, Socket.io, JWT, Multer | **Live:** eathub-swart.vercel.app | **Source:** github.com/goyalkartik773/Eathub

- Built a monolithic full-stack food delivery platform with three role-based dashboards (Customer, Restaurant Owner, Delivery Partner), each with distinct auth flows and real-time capabilities powered by Socket.io for live order updates without page refresh.
- Implemented dual authentication — Firebase Auth on the frontend combined with custom JWT middleware on the Express backend — securing all API routes and supporting role-based access control across the platform.
- Designed a normalized MongoDB schema (Users, Restaurants, Dishes, Orders) with Mongoose ODM, supporting image uploads via Multer and advanced dish filtering (veg/non-veg, price range, ratings) through optimized backend queries.

DPI Engine – Deep Packet Inspection & Network Traffic Analyzer — Dec 2025 – Present
Tech Stack: C++17, Python, PCAP, TCP/IP stack, TLS/SNI parsing, Multi-threading | **Source:** github.com/goyalkartik773/Deep-Packet-Inspection

- Built a multi-threaded C++ engine that parses raw PCAP captures layer-by-layer (Ethernet → IP → TCP/UDP → TLS), extracting SNI hostnames from TLS Client Hello handshakes to identify application-level traffic inside HTTPS flows.
- Designed a producer-consumer pipeline with 5-tuple hashing for stateful per-flow packet tracking and real-time blocking decisions; wrote Python scripts to generate test datasets and documented parsing logic as Markdown run-books.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

- **Academic Excellence:** Ranked in the **top 10%** of 500+ students in the Computer Science Engineering program at JIIT.
- **LeetCode:** Achieved **1700+ rating** with 600+ problems solved; earned 10+ achievement badges in Data Structures & Algorithms.
- **CodeChef & Codeforces:** 3-Star on CodeChef (global rank <1000) · Pupil on Codeforces (1350+ rating).
- **Google Big Code:** Qualified for online rounds, demonstrating strong algorithmic problem-solving and competitive programming skills.
- **Hackathon Finalist – Innovate 3.0:** Selected as a **finalist** among competing teams; built and presented a working project under time constraints — view project →